SEMINAR S1-7 (CO5151 – ADVANCED AGENTIC AI)

Agent Design Patterns & Orchestration

Workflows vs. Agents: Vơ sao kin trữc n gin nht thng lị la chn ững

Nhúm G4:

ng Lóm Tứng (Trng nhúm) · **Nguyn Trung Phong** · **V Vit Hng**

Ging viển hng dn: **TS. Lổ Xuón Bòch** | Ngiy bòo còo: 21/10/2026

Thi lng: 30 phữt + 15 phữt Q&A · 27 slides + 6 backup

Phón Cũng & Thi Lng Trơnh Bịy

Mi thịnh viổn ph tròch mt phn, kt hp lỳ thuyt vi demo vị s liu thc nghim

Phn I: Workflows & Còc Pattern C Bn

Nguyn Trung Phong

(8 phữt | Slides 5–9)

- Workflows vs. Agents
- Vơ sao Anthropic khuyổn bt u n gin
- Prompt Chaining & Routing
- Evaluator-Optimizer
- Khi nọo mi cn n agent

Phn II: Multi-Agent Orchestration

V Vit Hng

(8 phữt | Slides 10–15)

- Mũ hơnh Orchestrator-Workers (Anthropic)
- Orchestrator-Workers vs. long context
- Anthropic 2025: Lead/Subagent
- OpenAI 2025: tool loop, guardrails, evals
- LangGraph: cyclic state graph
- Durable execution & checkpointing

Phn III: Trade-offs & Kin Trữc ng Lóm Tứng (Trng nhúm)

(14 phữt | Slides 16–27)

- Swarm/Handoff & Human-in-the-Loop
- Scaling: bao nhiổu agent lị (Anthropic + Google 2026)
- Bng so sònh 5 kin trữc + ma trn 7 pattern
- Kin trữc cho bịi toòn G4
- ònh giò (Evals) + kt qu 10 ln chy
- Tng kt & Tho lun

Mc tiổu ca bui bòo còo

Tr li 7 mc bt buc trong cng + cóu defense, cú code demo vị s liu i kốm.

Thi lng: 30 phữt thuyt trơnh + 15 phữt Q&A (8 phữt phn bin + 7 phữt ging viổn); 6 backup slides.

K Hoch Np Bịi & Còc Mc Thi Gian

Còc mc chờnh vị 4 sn phm cn np

Còc mc thi gian

1. **Ch Nht, 18/10/2026 – 23:59**
   → **Hn np bịi** qua email.
   → Tiổu email: `[CO5151][G4] Seminar S1-7`.
2. **Th Hai & Th Ba (19–20/10)**
   → **Faculty audit window**: Thy xem slide vị ch ra nhng ch cha cht v k thut hoc s liu.
   → *Np tr s khũng nhn c pre-defense feedback.*
3. **Th T, 21/10/2026**
   → **Bòo còo chờnh thc trổn lp** (30 phữt).

4 Sn Phm Cn Np

1. `slides_s1_7.pdf`
   → 27 slides + 6 backup (bn ting Vit: `slides_s1_7_vi.pdf`).
2. `annotated_bibliography.pdf`
   → 1 trang: 1 core reading + 3 extension readings, mi bịi kốm 1 cóu critical reflection.
3. **Th mc `demo/` & code**
   → Web demo (`demo/web/`) vị script o token, latency, minh ha checkpoint recovery khi gp vn bn lut õ b thay th.
4. `ai_use_statement.pdf`
   → Bn khai bòo vic s dng AI theo quy nh ca trng.

Tịi Liu Tham Kho: Core & Extension Readings

1 core reading, 3 extension readings vị 1 tịi liu b sung 2025–2026

Core Reading (bt buc cho c lp)

Anthropic (Dec 2024): *Building Effective Agents* [Link]

- Phón bit **workflows** vị **agents**.
- Còc building block: prompt chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer.
- **Lun im chờnh:** *"When building applications with LLMs, we recommend finding the simplest solution possible, and only increasing complexity when needed."*

Extension Readings

1. **Anthropic (Jun 2025)**: *How We Built Our Multi-Agent Research System* [Blog]
   → Lead/subagent, context isolation; multi-agent tn khong 15× token so vi chat.
2. **OpenAI (Apr 2025)**: *A Practical Guide to Building Agents* [PDF]
   → Tool-calling loop, guardrails, evals, khi nịo nổn tòch thịnh nhiu agent.
3. **LangGraph Documentation (2024–2025)** [Docs]
   → Cyclic state graph, durable execution, checkpointing, human-in-the-loop.
4. **Google Research (Dec 2025; v3 Apr 2026)**: *Towards a Science of Scaling Agent Systems* [arXiv:2512.08296]
   → 260 cu hơnh: multi-agent thng task song song (+80,8%), thua task tun t (−70%).

Cóu Hi Trng Tóm Ca Seminar

Vơ sao Anthropic khuyổn chn gii phòp n gin nht, vị khi nịo thơ chi phờ token ca multi-agent orchestration (khong 15× so vi chat) lị òng b ra?

Workflows: Control Flow Nm Trong Code

- **Control flow:** vit sn bng code (Python, Go...).
- **Vai trò ca LLM:** thc hin tng bc nh trong mt pipeline õ nh trc.
- **Routing:** if-else, kim tra schema, chuyn trng thòi theo quy tc c nh.
- **u im:** d d oòn, d tòi lp, latency thp, ờt token (1×).

Agents: Model T Quyt nh Control Flow

- **Control flow:** do LLM quyt nh trong vùng lp (ReAct, tool calling).
- **Vai trò ca LLM:** t chn tool, quyt nh khi nịo dng vị i tip hng nịo.
- **Routing:** thay i theo observation runtime.
- **ònh i:** linh hot vi bịi toòn m, nhng li d cng dn vị chi phờ khú kim soòt.

nh Ngha Ca Anthropic (2024)

"Workflows are systems where LLMs and tools are orchestrated through predefined code paths. Agents, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks."

Vơ Sao Anthropic Khuyn u Tiến Workflow?

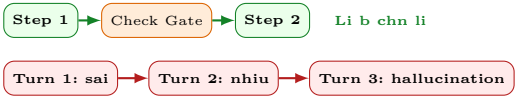
Thuyt trnh: Nguyn Trung Phong | Li cng dn ($P = p^n$), chi phờ token vị ri ro vn hịnh

Li Cng Dn Theo S Bc ($P = p^n$)

Gi nh c lp (i.i.d): $P = p^n$ (gi s $p = 0.95$ mi bc):

- **Workflow 3 bc:** $P_{\text{success}} = 0.95^3 \approx \mathbf{85.7\%}$
- **Agent loop 8 bc:** $P_{\text{success}} = 0.95^8 \approx \mathbf{66.3\%}$
- **Agent loop 15 bc:** $P_{\text{success}} = 0.95^{15} \approx \mathbf{46.3\%}$

Thc t trong Agent Loop: $P = p^n$ ch lị **cn trổn lc quan**. Do c ch t hi quy (autoregressive), li nh bc 1 góy **context drift / pollution**, khin xòc sut bc sau suy gim trm trng: $P(S_{k+1} \mid \neg S_k) \ll p$. Li thc t cng dn nhanh hn hịm m.



Bn Ri Ro Khi Trao Quyn T Ch Quò Sm

1. **Unbounded latency:** s turn do model quyt nh nổn thi gian phn hi khú d oòn.
2. **Token tng nhanh:** history dn lổn sau mi turn, tng token tng gn $O(N^2)$.
3. **Khú debug:** hịnh vi non-deterministic, rt khú tòi hin li edge case.
4. **Context drift:** tool output nhiu tờch t trong context, kỗo theo hallucination dóy chuyn.

Tr Li Cóu Hi

Vơ sao Anthropic khuyn khũng dứng agent cho nhiu bịi toòn? Vơ *"agentic systems often trade latency and cost for better task performance"*. Vi task õ rủ còc bc, workflow n nh hn, r hn vị d debug hn. Anthropic vit thng: *"This might mean not building agentic systems at all."*

Pattern C Bn: Prompt Chaining vị Routing

Thuyt trnh: Nguyn Trung Phong | Còch hot ng, li ởch vị còi giò phi tr

Pattern 1: Prompt Chaining



C ch: tòch task thịnh mt chui LLM call ni tip; output ca bc trc lị input ca bc sau.

c gở:

- Mi prompt ch lo mt vic, nổn d vit vị d kim soòt.
- Chốn c validation (gate) bng code gia còc bc.
- Còc bc trung gian cú th dứng model nh, r hn.

Mt gở:

- Latency cng dn theo s bc ($T = \sum t_i$).
- Cng nhc: mt bc sai mị khũng cú code r nhònh thơ li i thng xung còc bc sau.

Pattern 2: Routing



C ch: mt classifier (LLM, regex hoc embedding) xòc nh intent ri chuyn input sang nhònh x lỳ phứ hp.

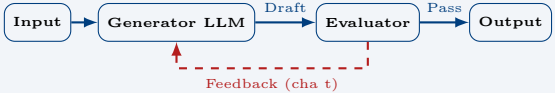
c gở:

- Mi nhònh cú system prompt vị b tool riổng, gn hn.
- Trònh prompt bloat (khũng phi nhi 50 tool vịo mt prompt).
- Query n gin cú th chuyn sang model nh tit kim.

Mt gở:

- Router phón loi sai thơ c nhònh phừa sau x lỳ sai theo.
- Thổm nhònh mi phi sa c router ln handler phừa sau.

C Ch Evaluator-Optimizer



Ba thịnh phn:

1. **Generator:** to output ban u t yổu cu.
2. **Evaluator (LLM hoc tool):** chm output theo tiổu chờ rủ rịng, tr v feedback cú cu trữc.
3. **Refinement loop:** generator sa theo feedback cho ti khi t yổu cu.

c gơ:

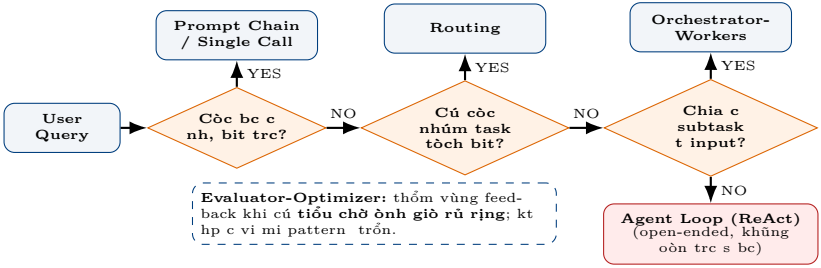
- Cht lng tng rủ code generation, dch thut, son vn bn.
- Ging còch con ngi vit nhòp ri nh ngi khòc review.

Chi Phờ & Còc Failure Mode Thng Gp

- **Token tng theo s vùng:** K vùng thơ tn khong $2K\times$ token (mi vùng 1 call generator + 1 call evaluator), latency cng tng theo.
- **Sycophancy:** nu evaluator dứng chung base model vi generator, nú d "gt u" vị b sút nhng li khú thy.
- **Diminishing returns:** cht lng thng tng nhiu nht vùng u; còc vùng sau d lùng vùng mị khũng tin b.
- **Còch khc phc:** dứng **external deterministic tools** (linter, unit test, API checker) lịm evaluator thay vơ LLM t review.

Khi Nịo Dứng Workflow, Khi Nịo Dứng Agent?

Thuyt trơnh: Nguyn Trung Phong | Còc cóu hi cn tr li trc khi chuyn t workflow sang agent



Gi Workflow n Gin Khi:

- ✓ Input/output schema õ rủ rịng.
- ✓ Failure mode chn c bng regex hoc Pydantic schema.
- ✓ Latency SLA cht (vờ d di 2 gióy), token budget hn ch.

Ch Dứng Orchestrator / Agent Khi:

- ▶ **Orchestrator:** subtask ch xòc nh c t input c th, cn **context isolation**.
- ▶ **Agent:** *"open-ended problems where it's difficult or impossible to predict the required number of steps"*.
- ▶ Kốm theo: sandbox testing, guardrails vị **checkpointing** cho task chy lóu.



Workflow Pipeline

- **Orchestrator:** Tip nhn input, phón tờch vị ch ng phón rõ nhim v thịnh còc subtask ng.
- **Worker LLMs:** Thc thi còc subtask c giao c lp / song song.
- **Synthesizer:** Thu thp kt qu t còc worker vị tng hp thịnh u ra hoịn chnh.

c Trng & Khi Nịo Nổn Òp Dng?

- **Linh hot:** Subtask khũng c nh trc mị c xòc nh da trổn input c th.
- **Bịi toòn phc tp:** Phứ hp cho nhim v khũng d oòn trc c danh sòch còc cũng vic con.
- **ng dng in hơnh:**
  - Cũng c lp trơnh cn thc hin thay i phc tp trổn nhiu file.
  - Nhim v tơm kim, thu thp vị phón tờch thũng tin t nhiu ngun.

Hn Ch Ca Single Long Context

- **Attention Dilution & Context Rot:** Nhi nhỗt còc ng cnh thũ (logs, tools, history) lịm tt dc t l Signal-to-Noise. Mũ hơnh d b sút rịng buc vị gp hin tng *Lost in the Middle*.
- **Truncation:** Khi chm trn ng cnh, c ch ct ta bt buc din ra, d lịm rng mt system prompt vị k hoch ban u.
- **Compounding Errors:** Li mt bc gi cũng c s b "neo"vịo chui t hi quy k tip, khin agent ri vịo vùng lp vũ tn hoc trũi khi mc tiổu.
- **Sequential Execution & High Cost:** Tin trơnh b khúa trong chui tun t $O(N)$; mi lt sinh token sau u chu chi phờ vị tr tờnh toòn trổn toịn b lch s thũ.

Li Th Ca Orchestrator-Workers

- **Context Isolation:** Mi subagent s hu context window riổng vị ch nhn ững cũng c chuyổn bit, **gim mnh** ri ro hallucinated tool arguments (khũng loi b hoịn toịn).
- **Information Compression:** Worker x lỳ hịng chc nghơn tokens thũ nhng ch bịn giao bn túm tt tinh gn cho Orchestrator, gi sch context cp cao.
- **Parallel Tooling:** Phón rõ bịi toòn m rng theo chiu ngang cho nhiu worker chy ng thi; trong h thng research ca Anthropic, còch nịy ct gim thi gian tơm kim ti 90%.
- **Compute Scaling & Model Tiering:** Cho phỗp tng token budget hiu dng vt trn n agent vị phi hp linh hot mũ hơnh ln vi mũ hơnh nh, r.

Kin Trữc & Ti u Nng Lc

- **Lead agent:** Gi research goal, phón rõ task, qun lỳ effort budget vị tng hp kt qu.
- **Subagents & Parallelism:** Chy song song (35 subagents, 3+ tool calls/ln) → gim 90% thi gian tơm kim.
- **Context hygiene:** Subagents ch gi v kt qu õ cũ ng, hoc xut ra Filesystem/Memory trònh "game of telephone".

Kt qu: Multi-agent (Opus 4 lead, Sonnet 4 subagent) t kt qu cao hn single-agent Opus 4 **90,2%** (tng i) trển internal eval.

ònh i & Thòch Thc K Thut

- **Token Cost:** Agent n tn ≈ 4×, multi-agent tn ≈ 15× token so vi chat (gii thòch 80% variance trển BrowseComp).
- **Latency Bottleneck:** Lead agent ch subagents ng b → 1 subagent b chm/treo s chn toịn b pipeline.
- **Compounding Errors:** Li nh tòch t trong quy trønh dịi. Cn Checkpoints, Diverse Sampling / Retries vị Fallback Logic khc phc.
- **o trển G4 (10 ln, ADK):** multi-agent 6,99 s vs ReAct 5,55 s (∼26% chm hn) nhng grounding ững; latency lị giò phi tr, khũng phi li.

Kt Lun Theo Cng

Multi-agent i token ly nng lc (chi phờ ≈ 15× **so vi chat**). Mũ hønh nịy ch hiu qu kinh t cho còc tòc v giò tr cao, cú tònh song song ln vị vt quò gii hn ca mt context window n l.

OpenAI: Agent Patterns, Guardrails & Evals

Thuyt trnh: V Vit Hng | Kin trũc thit k, cht chn an toịn vị phng phòp ònh giò

1. Single-Agent vs. Multi-Agent

- **Vùng lp thc thi:** Model iu khin chu trơnh `while` loop: nhn input → gi tool → nhn kt qu → dng khi t exit condition.
- **Ti u 1 agent trc:** Khuyn dứng prompt templates & policies trc khi tòch agent (ch tòch khi tool overload hoc logic quò r nhònh).
- **Manager vs Decentralized:** Manager gi subagents nh tool calls; Decentralized handoff ngang hịng gia còc agent.

2. Guardrails & Empirical Evals

- **H thng Guardrails bo v:**
  - *Input Classifiers:* Kim tra intent vị lc query c hi trc khi chuyn cho agent.
  - *Execution Bounds:* t trn s turn/tool call (Agents SDK: `max_turns`; LangGraph: `recursion_limit`) chn vùng lp vũ hn.
  - *Human Escalation:* Kờch hot cht chn ngi dứng cho hịnh vi ri ro cao.
- **Phng phòp Evals thc nghim:**
  - *Baseline First:* o t l hoịn thịnh tòc v trn workflow n gin trc khi dứng agent.
  - *Trajectory & Tool Checks:* o chờnh xòc chn tool vị i s Pydantic.
  - *Edge-case & Refusal Testing:* Kim tra kh nng t chi khi d liu móu thun/thiu.

Nguyn Tc Ct Lủi T OpenAI (2025)

"Our general recommendation is to maximize a single agent's capabilities first."
Thit lp b test baseline vị ònh giò li biển (edge-case failures) trc khi tng phc tp iu phi.

Cu Trữc Qun Lỳ State & Nodes

- **State Schema:** Cu trữc d liu dứng chung, lịm b nh ngn hn truyn qua toịn b th.
- **Reducers:** Hịm hp nht d liu (vờ d `operator.add` append message list).
- **Execution Nodes:** n v x lỳ c lpcú th lị LLM call, deterministic code, hoc mt Subagent.

Cyclic & Control Flow

- **Conditional Edges:** Hịm iu kin kim tra state nh hng lung.
- **Error Recovery & Retry Loop:** Cho phỗp agent th li hoc r nhònh sa li mị DAG khũng lịm c.
- **Bounded Loops:** Dứng `recursion_limit` vị retry counter kim soòt chi phờ, trònh lp vũ hn.

Giò Tr Ct Lủi Trong Orchestration

Dung hùa gia workflow c nh vị agent t do: LLM gi kh nng linh hot ra quyt nh, nhng lung thc thi vị chuyn trng thòi luũn cú cu trữc, kim soòt c vị d dịng kim toòn.

Durable Execution & Checkpointing: X Lỳ Li Trong Production

Thuyt trnh: V Vit Hng | State persistence, fault recovery vị human-in-the-loop

Durable Execution Lị Gơ?

- Agent chy task dịi **chc chn s gp s c**: rt mng, dờnh rate limit, container b kill.
- **Khũng cú checkpoint:** li bc 9 ngha lị chy li t bc 1, tn thổm c token ln thi gian.
- **Checkpointer ca LangGraph:** lu state xung SQLite/Postgres sau mi bc ca graph (state/checkpoint, khũng phi kho lut).
- **Fault recovery:** chy tip t checkpoint hp l gn nht.

Time Travel & Human-in-the-Loop

- **Interrupts:** graph tm dng trc còc hịnh ng ri ro cao.
- **Human review:** ngi kim tra state, chnh sa nu cn ri cho chy tip.
- **Time travel:** fork t mt checkpoint bt k trong quò kh th li hoc truy vt li.
- **Trong G4:** Human Gate bt buc trc `submit_portal_filing` (hịnh ng ghi ngoi vi khũng th hoịn tòc).

Bịi Hc Theo Cng

Durable execution bin mt agent prototype d v thịnh h thng cú th a vịo production, vơ phn vic õ lịm xong khũng b mt khi cú s c.

Pattern Agent-Level: Swarm & Handoff

Thuyt trnh: ng Lóm Tứng (Trng nhúm) | Khi còc agent ngang hịng t bịn giao hi thoi cho nhau

Swarm Lị Gơ?

- Nhiu agent **ngang hịng**, **khũng cú lead**; agent ang gi hi thoi t quyt nh **handoff** sang agent khòc.
- Khòc Orchestrator-Workers: worker khũng "bòo còo v lead" mị **thay phiổn** dn dt; state ch lị message history.
- **OpenAI Swarm (2024):** experimental, 2 primitive: `Agent` vị handoff; **stateless** (GitHub).
- Thay bng **OpenAI Agents SDK (2025)** — bn production (SDK Repo).

C Ch Handoff & Ri Ro

- Swarm: handoff lị hịm Python **tr v Agent**; mt lt gi nhiu handoff thơ **ch còi cui** cú hiu lc.
- Agents SDK: handoff i system prompt theo active agent nhng **gi nguyổn** history.
- Guardrail ch chy agent **u** vị agent **cui** ⇒ **on gia khũng c kim**; kốm ping-pong loop, message $O(N^2)$, khũng rollback.



Thc T: Triage Handoff vs. t Phò Gii Toòn Ca OpenAI

Tòc v chun (repo OpenAI Swarm): mt `triage_agent` iu hng sang `airline`, `refund`, `support`.
Chng minh hơnh thc (Lean 4): vi bịi toòn cn ũng/sai tuyt i, h thng chy **nhiu worker sinh chng minh c lp** ri **Lean kernel** kim tra — **khũng** dứng handoff/debate, vơ hi thoi t do gia nhiu agent n message $O(N^2)$ vị d to "o giòc ng thun".
G4 khũng chn Swarm: Cn audit *toịn b* lung vị chn *trc* hịnh ng ghi ra ngoịi — Swarm ch kim hai u chui.

Human-in-the-Loop: t Con Ngi Vịo ững im Dng

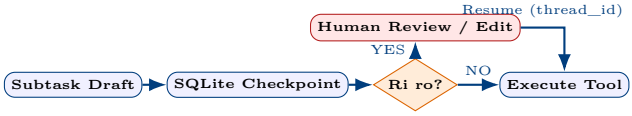
Thuyt trnh: ng Lóm Tứng (Trng nhúm) | Gate cp control flow: LangGraph interrupt() [Docs]

Ba Mc HITL (theo thi im can thip)

- **1. Hu kim (review output):** agent chy xong, ngi duyt. R nht, nhng li **õ xy ra**.
- **2. Tin kim / gate (approve action):** graph **dng trc** hịnh ng khũng hoịn tòc c. (G4 dứng mc niy).
- **3. ng kim (interrupt & edit):** ngi sa state gia chng; cn **checkpoint durable**.

Cịi t & ònh i

- LangGraph: `interrupt()` + SQLite checkpointer, resume cứng `thread_id` ⇒ quay li **ững node**.
- Gate theo **kh nng hoịn tòc** × **mc ri ro**.
- Gate quò nhiu ⇒ **rubber-stamping**, alert fatigue.
- Chi phờ: token gn nh khũng i, nhng **wall-clock tng**.

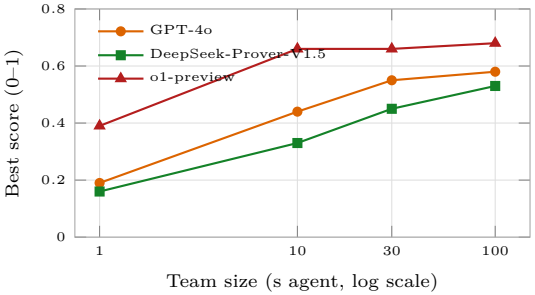


o Lng & Òp Dng Trong G4

Cóu hi phi tr li trc khi thổm gate: (1) hịnh ng niy cú hoịn tòc c khũng? (2) ngi duyt cú thũng tin quyt trong vịi gióy khũng? (3) nu ngi duyt sai thơ ai chu tròch nhim?
Metric: s ln interrupt trổn 100 task, thi gian ch trung bơnh, % quyt nh b ngi sa (quò thp ⇒ gate vũ dng; quò cao ⇒ agent cha tt).
Òp dng G4: HITL bc duyt k hoch subtask vị bc sinh hp ng cui — khũng gate tng cóu query SQL.

Scaling: Bao Nhiu Agent L ?

Thuyt trnh: ng Lóm Tng (Trng nhúm) | Bng chng t Lean theorem proving & t phò gii toàn ca OpenAI



Lean theorem proving: score theo team size (thang log). Ngun: DeepSeek-Prover-V1.5 / MiniF2F benchmark trn Lean 4.

[Lean 4 Prover]

c Biu & Quy Lut Knee

- **1 → 10 agent:** bc nhy ln nht (o1-preview: 0,39 → 0,66).
- **10 → 30:** Model mnh nht gn nh **i ngang** (0,66 → 0,66); hai dùng cùn li +0,11 v +0,12.
- **30 → 100:** Thm 70 agent ch +0,02; hai dùng kia vn +0,03 v +0,08.
- ⇒ **Bão hoà rõ nht model mnh (o1-preview), t 10–30 agent**; model yu vn tng chm.

Google 2026: Khi Nịo Multi-Agent Thng?

Thuyt trnh: ng Lóm Tứng (Trng nhúm) | 260 cu hnh, 6 benchmark [arXiv:2512.08296]

Thit K Nghiển Cu

- **260 cu hnh**, 5 kin trữc: **Single-Agent**, **Parallelization** (song song, khũng liển lc), **Orchestrator-Workers** (hub-and-spoke), **Swarm/Handoff** (peer-to-peer), **Hybrid** (orchestrator + peer).
- 6 benchmark: BrowseComp-Plus, Finance-Agent, PlanCraft, Workbench, SWE-bench Verified, Terminal-Bench — trển GPT-5, Gemini vị Claude Sonnet 4.x.
- "Agentic task" = multi-step vi mũi trng + thu thp thũng tin di **partial observability** + t iu chnh chin lc theo feedback.

Bn Kt Lun nh Lng

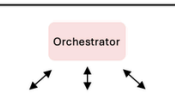
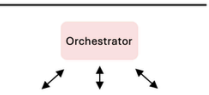
- **Alignment principle:** task **song song hoò c** (Finance-Agent) → **Orchestrator-Workers +80,8%** so vi Single-Agent.
- **Sequential penalty:** task **tun t** (PlanCraft) → *mi* bin th multi-agent u **gim 39,1–70,0%**: giao tip n ht "cognitive budget".
- **Tool-coordination trade-off:** cịng nhiu tool (16+), "thu" phi hp cịng ln.
- **Error amplification:** Parallelization **17,2×** vs Orchestrator-Workers **4,4×** — orchestrator lị **validation bottleneck**.

T Heuristic Sang K Thut: Mũ Hnh D Bòo Kin Trữc

Dứng **tool count** vị **decomposability** ca task d oòn kin trữc tt nht: $R^2 = 0{,}373$ (0,413 vi ACI), chn ũng cho **87%** cu hnh cha thy.
i chiu G4: pipeline G4 lị **chui tun t** (Planning → Retrieval → Audit → Drafting) nổn **khũng** hng +80,8%; còi G4 dứng lị **kim chng tp trung** (amp. 4,4× vs 17,2×).

Bng So Sònh 5 Kin Trữc Agent (Google, 2026)

Thuyt trơnh: ng Lóm Tứng (Trng nhúm) | Khung phón loi chònh thc: Interaction, Complexity, Overhead [Blog]

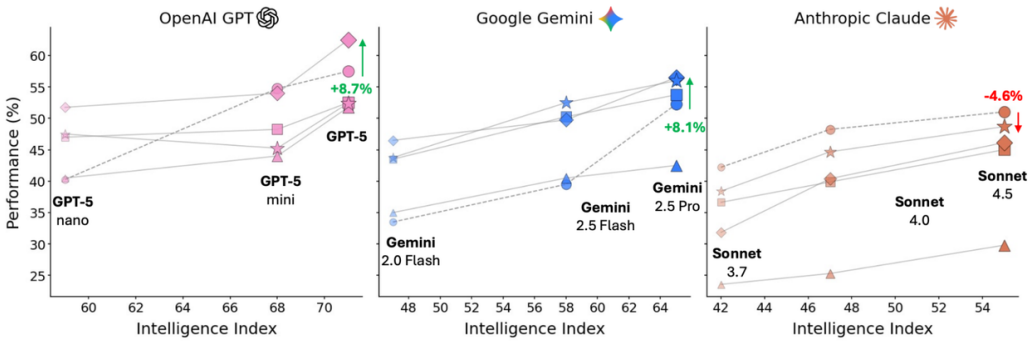
Characteristic	SAS	MAS (Independent)	MAS (Decentralized)	MAS (Centralized)	MAS (Hybrid)
Interaction Type					
LLM Calls	$O(k)$	$O(nk) + O(1)$	$O(dnk) + O(1)$	$O(rnk) + O(r)$	$O(rnk) + O(r) + O(p)$
Sequential Depth	k	k	d	r	r
Comm Overhead	0	1	$d \cdot n$	$r \cdot n$	$r \cdot n + p \cdot m$
Parallelization Factor	1	n	n	n	n
Memory Complexity	$O(k)$	$O(n \cdot k)$	$O(d \cdot n \cdot k)$	$O(r \cdot n \cdot k)$	$O(r \cdot n \cdot k + p \cdot n)$
Coordination	Sequential	Parallel + Synthesis	Sequential Debate	Hierarchical	Hierarchical + Peer
Consensus	-	Synthesis	Debate	Orchestrator	Orchestrator

Ngun: Google Research (Jan 2026), Towards a Science of Scaling Agent Systems, Research Blog & arXiv:2512.08296v3 (Table 2). k = max turns/agent, n = s agent, r = s vùng orchestrator, d = s vùng debate, p = s vùng peer.

i chiu vi la chn kin trữc ca G4: Mũ hơnh **Centralized (Orchestrator-Workers)** cú chi phờ LLM calls $O(rnk) + O(r)$ vị $r \cdot n$ overhead, nhng to ra **validation bottleneck** dp tt khuch i li t 17,2× xung cùn 4,4×. Do ú, G4 kiốn quyt chn Centralized (cịi trổn **Google ADK**, kốm checkpoint state SQLite) thay vđ Independent hay Swarm.

Hiu Nng Kin Trữc Theo 3 H Model (Google, 2026)

Thuyt trơnh: ng Lóm Tứng (Trng nhúm) | Bng chng thc nghim: The Myth of "More Agents" [Blog]



o lng trển 3 h model chờnh (OpenAI GPT, Google Gemini, Anthropic Claude) t Google Research (Jan 2026). Ngun: Research Blog & arXiv:2512.08296v3.

1. The Myth of "More Agents" & Capability Ceiling

- **OpenAI GPT (+8,7%) & Gemini (+8,1%):** Multi-agent vt tri Single-Agent khi model mc nng lc tm trung.
- **Anthropic Claude (-4,6%):** Khi chm ngng thũng minh cao (Sonnet 4.5), multi-agent **suy gim hiu nng** so vi Single-Agent (SAS)!

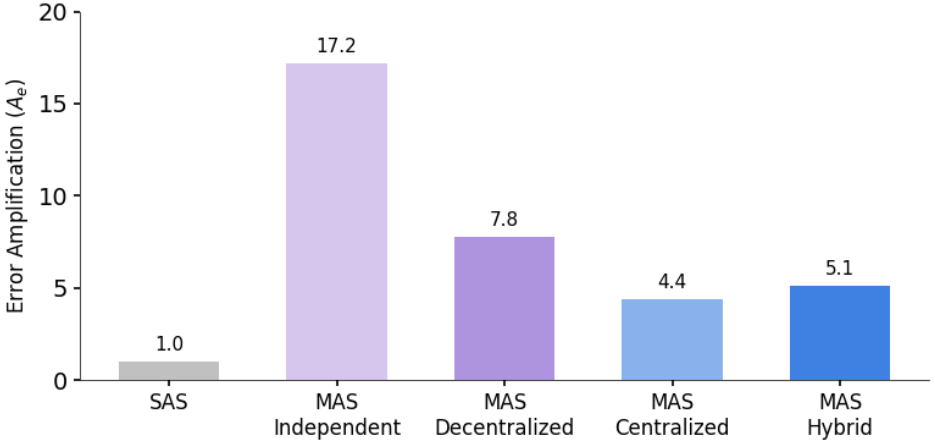
2. Ỳ Ngha Thit K Kin Trữc & La Chn G4

- **Nguyn lỳ ct lửi:** Khi baseline model õ mnh, thờm agent **khũng tng chờnh xòc** mị ch tng token, tr vị nhiu ng cnh.
- **ng dng G4:** Bounded autonomy — ch dứng 1 lead + 2–3 worker ti thiu cú kim chng, trònh ri vọ vứng suy gim hiu nng.

Li: Kin Trữc Nịo Khuch i Li?

Thuyt trơnh: ng Lóm Tứng (Trng nhúm) | Error taxonomy & error amplification (Google 2026, Bng 5 & Fig. 4)

[Blog]



Google Research (2026), Fig. 4 — Reliability: success rate & error containment theo kin trữc, trổn 3 h model (OpenAI GPT, Google Gemini, Anthropic Claude). Ngun: Research Blog & arXiv:2512.08296v3.

Ma Trn Trade-off: 7 Agent Design Patterns

*Thuyt trnh: ng Lóm Tứng (Trng nhúm) | Cứng b tiểu chờ: control flow, LLM calls, **memory**, latency, **risk***

[arXiv:2512.08296v3]

Pattern	Control Flow	LLM Calls	Memory	Latency	Risk	Dứng Khi
1. Prompt Chaining	Tuyn tờnh, ni tip	$O(k)$	$O(k)$ 1 context, tởch lu	k	Li bc trc lan sang sau; **khũng cú gate** ⇒ li ti thng output	Schema nh sn, khũng i qua còc ln chy
2. Routing	Phón nhònh, 1 nhònh	$O(1)+O(k)$	$O(k)$ 1 nhònh	1	Router phón loi sai ⇒ c nhònh sau sai; cha chm ngng coordination ($O < 100\%$)	iu hng model ln/nh theo intent
3. Parallelization	n nhònh song song, gp	$O(nk) + O(1)$	$O(nk)$ n context ri	k	$A_e^{trace} = \mathbf{17,2}\times$ [14,3; 20,1] — **cao nht**; gp khũng cross-validate ($O\% = 58\%$)	Cn a gúc nhơn, voting
4. Eval-Optimizer	Vùng lp trn K	$O(2K)$	$O(2K)$ 2 context/vùng	$2K$	Evaluator cứng h model ⇒ sycophancy; thiu verification tp trung thơ li khũng b chn	Sinh code, dch thut, son tho
5. Orch.-Workers	Lead phón rõ n worker	$O(rnk) + O(r)$	$O(rnk)$ gi $\sum$ summary	r	$A_e^{trace} = \mathbf{4,4}\times$ [3,8; 5,0] — kim li tt nht; nhng $O\% = 285\%$, E_c : 0,466 → 0,120; tool-heavy thơ $\beta_{E_c \times T} = -0{,}096$ ($p = 0{,}002$)	**Subtask cha bit trc, context isolation**
6. Swarm / Handoff	Ngang hịng, bịn giao	$O(dnk) + O(1)$	$O(dnk)$ mi agent gi	d	$A_e^{trace} = \mathbf{7,8}\times$; coordination failure 3,2%; $O\% = 263\%$; **khũng rollback** c state	Hi thoi a ch , sóu ngn
7. Human-in-the-Loop	Code dng, ngi duyt	$+O(1)$	$O(1)$ + checkpoint	+ ch	Rubber-stamping, alert fatigue; gate quò dịy thơ ngi duyt mt tòc dng	Hịnh ng khũng hoịn tòc c

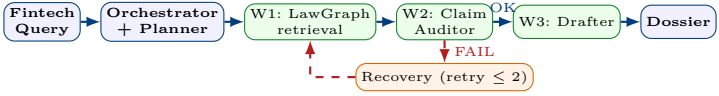
*Ngun: Anthropic (2024) cho pattern 1–5; OpenAI Swarm (2024) & Agents SDK (2025) cho 6; LangGraph cho 7. **Memory** vị **Latency** dứng ừng kỳ hiu Google (2026, Table 2): SAS $O(k)$ ề Independent $O(nk)$ ề Centralized $O(rnk)$ ề Decentralized $O(dnk)$; latency = sequential depth. **Risk:** A_e^{trace} = trace-level error amplification (Google, Table 5, $N = 260$): SAS 1,0× ề Independent 17,2× ề Decentralized 7,8× ề Centralized 4,4× ề Hybrid 5,1×; $O\%$ = coordination overhead so vi SAS. k bc/agent, n worker, r vùng orchestrator, d lt handoff, K vùng eval. **Ònh x khi i chiu:** Independent ≈ Parallelization, Centralized ≈*

Pattern B Loi: ReAct Loop T Do

- **Li tòch ly ($P = \prod p_i$):** Gi tool nhiu bc, xòc sut gim nhanh; li bc trc lịm lch context bc sau.
- **Bứng n token $O(N^2)$:** History dịi thổm sau mi turn vị gi li toịn b, tn 7.4× workflow.
- **Zero Fault Containment:** Khũng cũ lp c li gia còc bc; sai mt bc lị sai c chui. Trong case study G4, ReAct vn tr li ững nhng tn 7,4× token.
- **Loi c pipeline mt lt:** khũng cú bc kim tra hiu lc vị khũng quay li khi phòt hin vn bn õ **b thay th**.

Pattern c Chn: Orchestrator-Workers

- **Planner ng:** LLM planner tòch cóu hi thịnh **2 subgoal** + t khoò tra cu; s subgoal sinh theo tng cóu hi, khũng hard-code.
- **Context Isolation:** mi worker mt context riổng; Claim Auditor ch thy clause õ retrieve, khũng thy hi thoi.
- **Bounded Autonomy:** trn `max_retries` = 2 cho vùng recovery; cht chn kim tra hiu lc t CSDL tht.
- **Chy tun t:** 1 lead + 3 worker ni tip (Planning → Retrieval → Audit → Drafting), **khũng fan-out song song**.
- **Hiu qu thc t:** Tit kim 56% token so vi ReAct (case study, $n = 1$ query); grounding ững.



Chui tun t, mi worker mt context riổng, vùng recovery cú trn.

Bịi Hc Quyt nh Kin Trũc

Bt u t Simplicity-First. Nhng vi **zero-tolerance vi vn bn õ b thay th**, cn **planner ng + verification gate + recovery cú trn**. **Google (2026):** li ờch G4 hng lị **kim chng tp trung** (error amplification 4,4× thay vơ 17,2×),

Phng Phòp ònh Giò Agent (Evals & Benchmarks)

Thuyt trơnh: ng Lóm Tứng (Trng nhúm) | 6 nhúm ch s ngịnh dứng ònh giò agent

Nhúm	Ch S Hay Dứng	Tr Li Cóu Hi
1. Outcome	Task success rate; pass@k / pass^k; resolution rate (SWE-bench); accuracy / F1	Agent cú **lịm xong vic** khũng?
2. Trajectory	Tool-call accuracy; argument / schema validity; step count; plan adherence	Lịm **ững còch** khũng?
3. Retrieval (RAG)	Precision@k; Recall@k; Hit@k; MRR; nDCG@k	Ly **ững tịi liu** khũng?
4. Cost & Latency	Tokens/task; USD/task; p50 & p95 latency; cost-normalized accuracy	**Tn bao nhiổu** c im ú?
5. Reliability	Variance across runs; success@k; error amplification; recovery rate	Chy li cú **n nh** khũng?
6. Safety & Grounding	Grounding rate; hallucination rate; refusal rate; guardrail violations	Cú **ba** hoc **vt rịo** khũng?
7. Mi 2025–2026	Time horizon (METR); pass^k (τ-bench); cost-controlled accuracy; error amplification	Tr c **bao lóu**, **n nh**, **òng tin**?

Chn Ch S Nịo?

Outcome lị ch s chờnh; 6 nhúm cùn li lị **chn oòn**. Khũng bòo accuracy mị b cost (multi-agent $\approx 15\times$ token) $\Rightarrow$ phi bòo **cost-normalized accuracy**. **Time horizon** (METR, 2025–26) o bng *gi ngi* agent lịm c vi 50% reliability — thc o mi thay cho "im benchmark". **OpenAI (2025)** tòch *trajectory* vs *outcome* eval.

Vơ Sao Chn Kin Trữc Nịy? & Kt Qu Thc Nghim

Thuyt trơnh: ng Lóm Tứng (Trng nhúm) | 3 rịng buc rữt t case study ⇒ loi 3 kin trữc ⇒ o li trổn 10 ln chy

Rịng Buc Ca Bịi Toòn ⇒ Loi Kin Trữc Nịo

R1. Gp vn bn õ b thay th phi truy xut li

Tra ra N 101/2012 → tra field `superseded_by` trong graph (khũng phi LLM c lut) → thy 52/2024 thay th → phi ly iu mi. **Theo VBPL (B T phòp, 21/10/2026):** 101/2012 *cùn hiu lc n 01/07/2027* nhng õ b 52/2024 thay th — khũng phi "õ bõi b".

⇒ loi **Workflow tuyn tònh**: router–retriever–synthesizer, *khũng cú cnh quay lui.*

R2. Phi chn li trc khi ra output

Grounding sai lị li khũng hoịn tòc; LLM *t nguyn* gi tool kim hiu lc lị khũng .

⇒ loi **Swarm** (guardrail ch 2 u chui) vị **ReAct** (mt context, khũng gate bt buc).

R3. Token phi nm trong ngón sòch

ReAct gi li toịn b lch s mi bc: 66% input token ca nú lị observation lp li.

⇒ loi **ReAct** ln na: 7,4× workflow, 1,4× orch.-workers (ADK), 2,3× harness.

Giao Ca R1–R3 ⇒ Kin Trữc õ Chn

Orch.-Workers tun t (Planning → Retrieval → Audit → Drafting) + **recovery cú trn** (`max_retries`= 2) + **Human Gate**.

Google (2026, Table 5) o trổn *topology* ca h, khũng o h G4: tp trung hoò lị *validation bottleneck* — $A_e = 4{,}4\times$

o Thc T — 10 Ln/Chy, DeepSeek Flash V4.1

Kin Trữc	Tokens (Mean ± Std)	Latency
1. Workflow	849 ± 11 (1,0×)	3,20 ± 0,23 s
2. Monolithic ReAct	6,313 ± 125 (7,4×)	5,55 ± 0,51 s
3. Multi-agent (harness LangGraph)	2,760 ± 513 (3,3×)	7,44 ± 1,76 s
4. Multi-agent (Google ADK — G4)	4,495 ± 224 (5,3×)	6,99 ± 1,03 s

*Cứng 1 query, c 4 u tr li ừng. **Hịng 3** = harness vit li bng LangGraph (2 LLM call, audit bng code Python); **hịng 4** = pipeline production `adk_pipeline.py` (SequentialAgent + LoopAgent, state SQLite), tr **nguyn iu** 6 iu/lt nn tn hn 1,6× mị cn c hn. Trn 100 cu SBV, c 4 dng cng mt retriever dense: R@k = 97 / 99 / 95 / 97%; giòm kho chm "correct" 46,0 / 44,0 / 41,0 / 40,0%. Chi tit: Backup 4.*

ònh i õ Chp Nhn — Vơ Sao Vn Chn Hịng 4

Chm hn Workflow 2,2×, tn 5,3× token. Trn 100 cu SBV tng quòt Workflow cùn ững *hn* (46,0% vs 40,0%) — ngha lị kin trữc nịy **khũng** mua bng accuracy trung bnh.

Nú mua **ng zero-tolerance**: cu hi cú vn bn õ **b thay th** thơ workflow tuyn tònh *khũng th* quay lui, cùn ReAct ch ững khi LLM tnh c gi ững tool. i li c **faithful 94%** vị bòm gii hn 150 t (median 102 t).

Tr li thng nu b hi: trổn bịi toòn ph thũng thơ workflow thng; G4 chn orch.-workers cho ng audit, khũng phi tng im trung bnh.

Tng Kt & Tho Lun

Còc kt lun chònh ca Seminar S1-7

Thũng ip Chònh:

Agent linh hot nhng d v; vi task õ rủ còc bc, workflow n nh vị r hn.
Bng chng nh lng 2026 (260 cu hnh, Google Research): **kin trữc phi khp cu trữc task** — song song hoò c thơ Orchestrator-Workers thng +80,8%, tun t thơ multi-agent **gim** 39–70%.
Khi bịi toòn buc phi iu phi ng, **lead/subagent vi context isolation** kt hp **c ch checkpointing bn vng (theo nguyổn lỳ LangGraph)** lị hng i an toịn nht a vịo production.

Cm n Thy Vị Còc Bn õ Lng Nghe!

Nhúm G4 rt mong nhn c cóu hi vị gúp ỳ t TS. Lổ Xuón Bòch vị còc bn.

Nhúm G4: ng Lóm Tứng (Trng nhúm) · Nguyn Trung Phong · V Vit Hng | CO5151 - HK261

Backup: Parallelization — Sectioning vị Voting

Dứng khi còc nhònh c lp: tòch vic chy ng thi, hoc ly nhiu mu b phiu

1. Sectioning (chia vic)

- Tòch task thịnh còc nhònh c lp, chy ng thi, ri gp kt qu bng code.
- Phứ hp: nhiu guardrail kim tra song song; x lỳ nhiu tịi liu hoc nhiu h s cứng lữc.
- Chi phờ: N nhònh $\rightarrow N\times$ token; latency bng nhònh chm nht ($\max t_i$).

2. Voting (b phiu)

- Chy cứng mt prompt nhiu ln (hoc nhiu model), tng hp bng b phiu hoc ngng.
- Phứ hp: phón loi, kim tra tờnh ững n khi sai sút cú chi phờ cao.
- Chi phờ: $N\times$ token; i li gim variance vị tng tin cy.

Khi nịo KHÕNG dứng

- Còc bc ph thuc nhau (output bc A lị input bc B) $\rightarrow$ phi tun t.
- Ngón sòch token/latency cht: song song khũng gim token, ch gim wall-clock.
- Chi phờ gp kt qu ln hn li ờch (nhiu nhònh, tng hp phc tp).

Liổn h G4

H thng G4 hin chy retrieval $\rightarrow$ audit $\rightarrow$ drafting **tun t** vơ auditor ph thuc kt qu retrieval. Hng m rng: chy song song còc subgoal c lp vị nhiu guardrail.

Backup: Bn 7 Mc Bt Buc & Q&A Nhanh

Dứng khi b hi li: mi mc bt buc nm slide nọ, vị 3 phn bin thng gp

Mc bt buc → Slide

1. Workflow vs. agent: code path vs. model-directed → **Slide 5**
2. ≥ 3 pattern + buys/costs → **Slide 7–8, 10, 16–23**
3. Vơ sao simplest + khi nọ nóng cp → **Slide 6, 9**
4. Durable execution / checkpointing → **Slide 15 + Backup 5**
5. Lead/subagent i cost & latency → **Slide 11–12**
6. Quyt nh ca G4 + alternative b loi → **Slide 24**
7. Orchestrator-worker vs. one long context → **Slide 11**

Defense: vơ sao Anthropic chng agent cho nhiu task? → **Slide 6**

3 Cóu Phn Bin Thng Gp

- **Vơ sao va khuyổn simplicity-first va xóy orchestrator?** Workflow b sút iu 22 ca N 52/2024 khi vn bn c (101/2012) vn c tr v vị *vn cùn hiu lc n 2027*; orchestrator cú bounded autonomy lị mc phc tp ti thiu grounding ững.
- **Vơ sao cn External Feedback?** Self-correction khũng cú phn hi t cũng c/CSDL bổn ngoịi thng khũng sa c li; cn tra cu CSDL thc t bit vn bn õ ht hiu lc.
- **Durable execution hot ng th nọ?** Checkpoint ti còc node trng yu (C1/C2/C3) xung SQLite (state; kho lut Neo4j/Qdrant); li bc 3 thơ resume t checkpoint gn nht, khũng cold restart.

Backup: Ranh Gii T Ch — Tool Permissions & Human Gate

Security óy lị design pattern ca orchestration: ai c gi tool nịo, guardrail t óu

Ai c lịm gơ?

- **Orchestrator:** lp k hoch, iu phi, ghi state.
- **LawGraphAgent:** ch c (retrieval).
- **ClaimAuditor:** ch c (validity check), khũng cú tool ghi.
- **Drafter:** tng hp, khũng gi tool ngoi vi.
- **`submit_portal_filing`:** khũng agent nịo gi c; bt buc Human Token.

Vơ sao lị design pattern?

Bounded autonomy + least privilege + human-in-the-loop: bin "agent t do" thịnh "agent cú quyn hn hp", ững tinh thn Simplicity-First.

Guardrail t óu?

- **Trc tool call:** kim tra schema + quyn ca agent.
- **Sau tool output:** validate d liu tr v.
- **Trc hịnh ng khũng th hoịn tòc:** Human Gate (token mt ln, ht hn 300 s).
- **Auditor isolation:** ch thy candidate clauses, khũng thy ỳ nh ngi dứng.

Map vi readings

OpenAI (2025): guardrails + human intervention cho high-risk actions. OWASP LLM06 (2025): Excessive Agency (bn 2023–24 gi lị LLM08). Anthropic (2024): ch tng autonomy khi tht s cn.

Backup: o Lng Orchestration — Trajectory, Outcome, Cost

Evals khũng ch chm im cui; nú lị tờn hiu quyt nh cú cn nóng cp kin trữc hay khũng

3 tng o

1. **Trajectory:** chn ũng tool, ũng schema, ũng th t handoff.
2. **Outcome:** kt qu cui ũng lut, ũng iu khon, grounding ũng.
3. **System:** token, latency, recovery sau checkpoint.

Quy tc quyt nh

Ch nóng lổn multi-agent khi **trajectory eval cho thy workflow khũng** (cn phón rõ ng hoc context isolation), vị khi **cost/latency** vn trong SLA. Nu khũng, gi workflow.

100 cóu SBV × 4 kin trữc (cứng 1 retriever dense)

Kin trữc	Tokens	R@k	Judge
Workflow	957	97%	46,0%
ReAct	5.596	99%	44,0%
Multi-agent (LangGraph)	1.540	95%	41,0%
Multi-agent (ADK)	2.394	97%	40,0%

Judge = % cóu c LLM giòm kho chm "correct" so vi `reference_answer` cú sn. R@k gn bng nhau ⇒ khòc bit n t **kin trữc**, khũng t truy xut.

Chi Phờ c 1 Cóu ũng

Tokens cho 1 cóu ũng: Workflow **2.081** · LangGraph 3.756 · ADK 5.984 · ReAct 12.719. C ba cu hnh multi-agent u **t hn mị khũng ũng hn** workflow — lun im Simplicity-First cú s liu. Judge thng nht vi ngi soòt 19/20 mu.

Backup: State & Memory Contract — Ai Thy Gơ, Lu Gơ

Context isolation khũng ch lị "chia nh prompt"; nú lị hp ng state gia còc agent

3 tng state

- **Working memory:** `LegalAgentState` (query, plan, clauses, audit report).
- **Durable state:** checkpoint C1/C2/C3 (SQLite/Postgres) state, khũng phi kho lut.
- **Static KB:** Neo4j (graph lut) + Qdrant (vector), ch c.

Checkpoint & rollback

- C1 sau planning, C2 sau retrieval, C3 sau dossier.
- Li C2 → rollback C1, gi subgoal hp l, ch chy li phn sai; retry ≤ 2.

Context isolation contract

- **Orchestrator:** thy plan + summary, khũng thy raw scratchpad.
- **Worker:** ch thy subgoal + tool ca mơnh.
- **Auditor:** ch thy candidate clauses, khũng thy ỳ nh ngi dứng.
- **Drafter:** ch thy clause õ verify.

Vơ sao quan trng

Gim attention dilution, chn li lan sang node khòc, vị bin **memory** thịnh th kim toòn c thay vơ mt context khng l.

Backup: So Sònh Còc Topology — Hub-and-Spoke, Swarm & Cyclic Graph

Dứng khi Q&A hi v kin trữc giao tip: ba còch t chc lung iu phi (Swarm y Slide 16, bng 5 kin trữc Slide 20)

1. Hub-and-Spoke (Anthropic)

Mũ hơnh tp trung:

- Lead iu phi; subagent lịm vic c lp ri bòo còo li cho lead.
- **u im:** lead nm toịn b state; context ca còc subagent tòch bit.
- **ònh i:** latency ph thuc subagent chm nht (**straggler effect**, $\max t_i$); tn nhiu token (khong 15× chat).

2. Decentralized Swarm

Peer-to-peer:

- Agent t handoff vị trao i trc tip vi nhau (vờ d: OpenAI Swarm, tin thón ca OpenAI Agents SDK).
- **u im:** linh hot vi hi thoi nhiu bổn, khũng cn cu trữc c nh.
- **Ri ro:** s message cú th tng $O(N^2)$; d kt vùng lp ping-pong; khú rollback state.

3. Cyclic State Graph

LangGraph:

- iu phi qua state graph vi typed schema vị conditional edges.
- **u im:** kim soòt c vùng lp; cú sn checkpointing vị time travel.
- **ònh i:** phi thit k state schema cn thn t u; kỗm linh hot hn swarm.

La Chn Cho H Thng Production

H thng cn tin cy cao nổn trònh swarm khũng cú rịng buc. Hng ph bin hin nay lị **hub-and-spoke phón cũng vic**, kt hp vi **cyclic state graph x lỳ li vị recovery**. **i chiu thut ng vi paper Google (2026):** SAS = Single-Agent; Independent = Parallelization; Centralized = Orchestrator-Workers; Decentralized = Swarm/Handoff; Hybrid = Hybrid (orchestrator + peer, tc hub-and-spoke cú thổm peer link). Error amplification: 4,4× (Orchestrator-Workers) vs 17,2× (Parallelization).